

Artificial intelligence for diagnostics in radiology practice: a rapid systematic scoping review



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Summary

Background The aim of this review was to evaluate evidence on the use of Artificial Intelligence (AI) to support diagnostics in radiology, including implementation, experiences, perceptions, quantitative, and cost outcomes.

Methods We conducted a systematic scoping review (PROSPERO registration: CRD42024537518) and discussed emerging findings with relevant stakeholders (radiology staff, public members) using workshops. We searched four databases and the grey literature for articles published between 1st January 2020 and 31st January 2025. Articles were screened for eligibility ($N = 8013$), resulting in 140 included studies. Studies evaluated implementation ($N = 7$), perceptions ($N = 74$), experiences ($N = 14$), effectiveness ($N = 53$), and cost ($N = 6$).

Findings Factors influencing AI adoption were identified, including the high technical demand, lack of guidance, training/knowledge, transparency, and expert engagement. Evidence demonstrated improvements in diagnostic accuracy and reductions in interpretation time. However, evidence was mixed regarding experiences of using AI, the risk of increasing false positives, and the wider impact of AI on workflow efficiency and cost-effectiveness.

Interpretation The potential benefits of AI are evident, but there is a paucity of evidence in real-world settings, supporting cautiousness in how AI is perceived (e.g., as a complementary tool, not a solution). We outline wider implications for policy and practice and summarise evidence gaps.

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Introduction

The use of Artificial Intelligence (AI) in healthcare is increasing globally due to its potential to address workforce shortages and rising healthcare demands.^{1–3} In radiology imaging, AI is being applied to assist with detecting abnormalities, enhance accuracy, reduce routine task time, and support clinical diagnoses.^{4–6}

However, evidence of these potential impacts is inconsistent, making it challenging to draw conclusions.⁶ While there is excitement and optimism about the use of AI, there is limited research evaluating the effectiveness of AI in real-world healthcare settings, which goes beyond testing how AI could theoretically work.^{2,6,7}

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Research in context

Evidence before this study

There is limited existing knowledge about how Artificial Intelligence (AI) is being implemented in radiology settings and the clinical implications of using these technologies. Existing reviews on the use of AI for diagnostics in radiology have not yet evaluated literature which integrates findings on: implementation; perceptions of staff, patients and the public; staff experiences; and impact, including effectiveness and cost.

Added value of this study

This review summarises current evidence on how AI is being implemented in diagnostic imaging, what staff, trainees, patients, and members of the public think about AI, experiences of using AI in practice, the quantitative impact, and cost. Furthermore, we build on existing work by

discussing review findings with staff in radiology and members of the public who may experience AI-based care using workshops.

Implications of all the available evidence

Research suggests that current AI implementation is based on experimental learning rather than being informed by rigorous evidence. To understand how to best use AI in the complexities of radiology practice, we highlight the importance of evaluating how AI is implemented and used as a complementary tool in real-world settings. Future research should focus on key evidence gaps, including the process of implementation (including procurement), experiences of using AI in practice, long term cost-effectiveness, the risk of increasing false positives, and the impact on wider patient pathways and hospital systems.

Recent guidelines for clinical implementation highlight the importance of addressing evidence gaps, emphasising that a strong evidence base is essential for the continued use of AI in radiology.⁸ With interest in AI on the rise, it is crucial to understand how AI is being adopted globally for diagnostic purposes, evaluate the overarching effectiveness and costs of these technologies, and consider the experiences and perceptions of relevant stakeholders. Existing reviews have not yet evaluated literature integrating all of these topics. To address this gap, we conducted a rapid systematic scoping review and stakeholder workshops. Our review addresses the following research questions:

1. How have AI tools been implemented to support diagnostics in radiology?
2. How have AI tools supporting diagnostics in radiology been experienced and perceived?
3. What evidence exists on effectiveness and cost of AI tools to support diagnostics in radiology?
4. How has evidence on implementation, experiences, perceptions, quantitative outcomes, and cost of AI been measured, collected, and analysed?
5. What do stakeholders (staff and the public) think about the review findings?

Methods

Registration

This review was registered on PROSPERO (CRD42024537518).

Design

As part of a wider rapid evaluation of AI deployment for chest diagnostics in the English National Health Service (NHS) (part of the AI diagnostic fund, AIDF),⁹ we conducted a rapid systematic scoping review.¹⁰ We used the

Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) statement,¹¹ supplemented by two stakeholder workshops (one with radiology staff and one with members of the public) to discuss review findings. Scoping review methodology was chosen to address the varied review topics in a field of emerging evidence¹² and involves bringing together evidence and summarising the findings.¹³ In rapid scoping reviews, the scope/focus and processes are streamlined to enable a quicker evidence synthesis.¹³

Search strategy and selection criteria

Studies eligible for inclusion:

- Focused on AI being used to support diagnostics in radiology (algorithmic use for image interpretation and decision-making, not image generation).
- Empirical studies (covering implementation, experiences, perceptions, quantitative or cost outcomes). Quantitative studies needed to evaluate AI as a support tool for human decision-making, rather than being used in isolation; in line with current guidance that AI should be used with human supervision.^{14,15}
- Published between 1st January 2020 and 31st January 2025, due to the rapid advancements of technology within healthcare during this period, and the publication of policy documents on the use of AI.¹⁵
- Covered United Kingdom (UK)-based and international evidence.
- Written in English.

See [Appendix S1](#) for detailed inclusion and exclusion criteria.

Four databases were searched: Medline-Ovid [PubMed], PsycInfo, CINAHL Plus, and Web of Science Core Collection. Grey literature was identified through policy websites, topic-specific websites, and

grey literature databases (e.g., Royal College of Radiologists, The Health Foundation, and Google Scholar). We also searched reference lists of review articles^{2,16,17} and discussed identification of further papers with external stakeholders and clinical experts (TOR, FG). One researcher (RL) conducted the search and inputted records onto EndNote,¹⁸ followed by Rayyan,¹⁹ to remove duplicates.

The search strategy was developed iteratively using Ovid-Medline, with assistance from a UCL librarian (DM). Search terms were informed by previous research¹⁶ and guidance from clinical experts (TOR, FG). These terms covered areas such as AI, radiology imaging, clinical practice implementation, experiences and/or perceptions and quantitative and cost outcomes (Appendix S2).

Screening and study selection

Studies were screened in three phases: i) title, ii) abstracts/study summaries, and iii) full texts.¹⁰ One researcher (RL) screened all titles and abstracts. 10% of excluded papers were reviewed by one of three researchers (ED, EM, CSJ). Full texts were screened by one of four researchers (RL, ED, EM, CSJ), depending on the methodology of the paper. Disagreements were discussed within the team until a consensus was reached.

Data extraction and charting

A data extraction tool was developed. This included study characteristics, setting, implementation context, methods and design, and study findings pertaining to 1) implementation, 2) experiences, 3) perceptions, and 4) quantitative and cost outcomes (Appendix S3). One of four researchers (RL, ED, EM, CSJ) piloted the tool on one study relating to each of the different topics explored in the review. The data extraction tool was used to extract findings from all studies. Disagreements when developing and editing the tool were discussed within the team until a consensus was reached.

Synthesis of results

Narrative synthesis was used to analyse review findings.²⁰ We extracted all data relating to implementation, experiences, and perceptions, including illustrative quotes from qualitative studies. Data were coded line-by-line and findings grouped thematically. Quantitative evidence on effectiveness was synthesised by organising results by diagnostic accuracy, time and workflow, and change in clinical decision making. The narrative synthesis for cost outcomes was supported by abridged data extraction tables, including incremental costs, incremental cost-effectiveness ratio, cost savings, net present value, and quality-adjusted life years (QALYs). Due to the heterogeneity of the studies, a quantitative synthesis was not feasible.

Critical appraisal

The Mixed Methods Appraisal Tool²¹ was used to evaluate study quality. The tool was applied to quantitative

effectiveness studies by two researchers (ED, CSJ), qualitative studies, mixed-methods, and quantitative survey studies by one researcher (RL). The quality of cost studies was evaluated (EM) using the Drummond checklist²² and the rating scale developed by Doran.²³

Stakeholder workshops

Two online stakeholder workshops were conducted. Workshops were held with members of the public and radiology staff working in an English healthcare setting (Appendix S4). Participants were recruited via public involvement in research channels, or via AIDF colleagues and clinical expert co-authors sharing the advert with radiology staff via email. Participants were selected based upon their experience in radiology and/or their experiences of diagnostic care, to ensure that a range of perspectives were included (Appendix S4). All participants were sent an information sheet and consent form ahead of the workshop. Preliminary findings were sent to participants ahead of the workshop and presented during the workshop (Appendix S5). Participants then discussed the findings. Workshops were audio-recorded and transcribed. Findings were analysed using thematic analysis²⁴ structured around review findings.

Ethics

Ethical approval for the workshops was obtained from the University College London Ethics Committee (27037/001). Written informed consent was obtained from all participants before taking part in the workshops. Ethical approval was not required for the review.

Role of funding source

The funders did not have a role in study design, data collection and analysis, writing of the manuscript or the decision to publish. The views and opinions expressed are those of the authors and do not necessarily reflect those of the NIHR or the Department of Health and Social Care.

PPIE co-authors (RM, JL, AH) were involved in study conceptualisation and design. They also co-designed study materials for the workshops, including the summary document sent to participants and presentation slides.

Results

Study selection and characteristics

8013 studies were identified, and 140 studies included (see Fig. 1), of which 7 studied implementation, 14 experiences, 74 perceptions, 53 quantitative impact and effectiveness, and 6 cost-effectiveness. Forty of the included studies were published in 2024–January 2025, indicating rapid growth in the AI research field ($N = 13$ published in 2020, $N = 27$ in 2021, $N = 25$ in 2022, $N = 35$ in 2023, $N = 38$ in 2024, and $N = 2$ in January 2025). Some studies covered multiple topics and have been included more than once in Table 1. Included

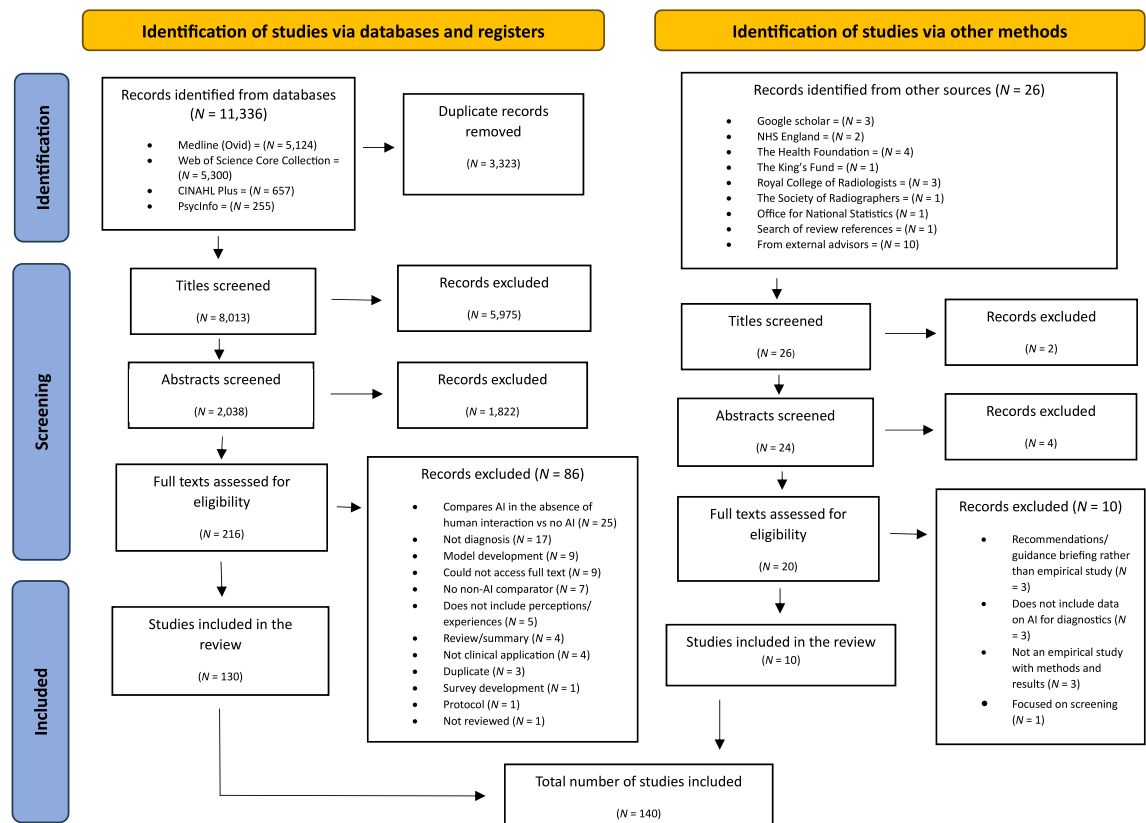


Fig. 1: Preferred reporting items for systematic reviews and meta-analyses (PRISMA) flow diagram.

studies were conducted in varied countries, covered different imaging modalities, and applied AI to varied patient pathways (Table 1).

Methods used for data collection and analysis

All implementation studies (N = 7) used qualitative methods (see Appendix S6). Experiences (N = 14) and perceptions (N = 74) were evaluated using quantitative, qualitative, and mixed methods. Of the quantitative studies that measured effectiveness of AI (N = 53), only 23 studies^{25–47} evaluated AI in a live pathway, of which 7 measured diagnostic accuracy. The remaining studies simulated typical workflow, often using a bespoke tool developed and validated as part of the study or a companion study. The effect of reader experience was the most common influencing factor measured to assess effect on findings (25/53).

The cost-related studies (N = 6) used Markov based models^{37,38,48–50} and Monte Carlo forecasting methods⁵¹ to demonstrate the costs and health outcomes of AI-aided and non-AI strategies. Analyses were conducted from a societal perspective in the context of the origin country, using observational data and assumptions as well as parameters from randomised trials. Costs were estimated using the origin country's public and private reference costs.

The AI tools used in quantitative studies of effectiveness were either a pre-existing commercially available tool or a bespoke tool designed as part of the study or companion study (Appendix S6 and Supplementary File S1).

Measuring effectiveness and cost effectiveness

Quantitative measures used to evaluate effectiveness covered three categories: (1) Diagnostic accuracy, (2) effect on time and workflow, and (3) effect on clinical decision-making. General measures of diagnostic accuracy included sensitivity, specificity, area under the curve (AUC), positive predictive value (PPV), negative predictive value (NPV), and accuracy (proportion of correct results). Cost-effectiveness studies measured health outcomes including tooth retention time (in years) until carries detection for recipients of oral health care and quality-adjusted life years (QALYs). Other outcomes measured included incidence, mortality, scan time, and withdrawal time (Supplementary File S1).

Implementation

Out of the seven implementation studies, four explored the process of implementation, including the integration of AI into clinical practice and associated barriers and facilitators.^{52–55} Three studies^{56–58} explored experiences of

Study characteristics		Overall (N = 140)				
Study focus ^a						
Implementation		7				
Experience		14				
Perceptions		74				
Effectiveness (quantitative studies)		53				
Cost		6				
Focus across multiple topics ^a						
Implementation, experiences, and perceptions		3				
Experiences and perceptions		3				
Implementation and perceptions		1				
Effectiveness and cost effectiveness		2				
Effectiveness and experiences		2				
Study characteristics	Overall (N = 140)	Study focus ^a				
		Implementation (N = 7)	Experiences (N = 14)	Perceptions (N = 74)	Effectiveness (quantitative) (N = 53)	Cost (N = 6)
Location						
United Kingdom (UK)	17	4	2	13	2	1
China	16	0	1	3	13	0
United States of America (USA)	15	0	3	5	8	2
Korea	11	0	3	1	8	0
Germany	9	0	1	5	3	1
Saudi Arabia	9	0	0	9	0	0
Italy	7	0	1	4	3	0
The Netherlands	6	3	1	2	2	0
Australia	5	0	0	3	2	0
India	3	0	0	2	1	0
Japan	3	0	0	0	2	1
Spain	3	0	0	3	0	0
World-wide	3	0	0	3	0	0
Africa	2	0	0	2	0	0
Europe-wide	2	0	1	1	0	0
France	2	0	0	0	2	0
Nordic countries	2	0	0	2	0	0
Singapore	2	0	0	1	1	0
Switzerland	2	0	0	0	2	0
United Arab Emirates	2	0	0	2	0	0
Argentina	1	0	1	0	0	0
Australia & New Zealand	1	0	0	1	0	0
Austria	1	0	0	1	0	0
Canada	1	0	0	1	0	0
China and Germany	1	0	0	0	1	0
Egypt	1	0	0	0	1	0
Finland	1	0	0	0	0	1
Ghana	1	0	0	1	0	0
Ireland	1	0	0	1	0	0
Jordan	1	0	0	1	0	0
Malaysia	1	0	0	1	0	0
Malta	1	0	0	1	0	0
Middle East and India	1	0	0	1	0	0
Nigeria	1	0	0	1	0	0
Taiwan	1	0	0	0	1	0
Thailand	1	0	0	1	0	0
Unclear	1	0	0	1	0	0
Vietnam	1	0	0	0	1	0
Western Europe	1	0	0	1	0	0

(Table 1 continues on next page)

Study characteristics	Overall (N = 140)	Study focus ^a				
		Implementation (N = 7)	Experiences (N = 14)	Perceptions (N = 74)	Effectiveness (quantitative) (N = 53)	Cost (N = 6)
(Continued from previous page)						
Imaging modalities						
Any (not specified—focused on AI more broadly)	67	3	3	65	0	0
X-ray	19	1	2	1	16	1
Computed tomography (CT)	18	1	2	2	16	0
Magnetic resonance Imaging (MRI)	10	1	1	2	7	1
Colonoscopy	7	0	1	1	5	2
Ultrasound	7	0	0	0	7	0
Radiographs	5	0	5	1	0	0
Mammography	2	0	0	0	2	0
MRI/MRI fusion biopsy	1	0	0	1	0	0
Computed tomography (CT) or magnetic resonance imaging (MRI)	1	0	0	1	0	0
Computed tomography angiography (CTA)	1	0	0	0	0	1
Images acquired during endoscopic procedures	1	0	0	0	0	1
X-ray, CT, DXA (bone density scan), and MRI	1	1	0	0	0	0
Patient pathways (including condition/s)						
Any (not specified—focused on AI more broadly)	66	3	3	63	0	1
Chest/thoracic/lung	23	1	5	3	17	0
Colorectal	7	0	1	1	5	2
Prostate	7	1	1	3	4	0
Fractures	5	0	2	0	4	0
Stroke	5	0	0	0	4	1
Breast	4	0	0	0	4	0
Coronary artery disease	3	0	0	0	3	0
Pulmonary embolism	3	0	0	0	3	0
Thyroid	3	0	0	0	3	0
Dentistry	2	0	0	1	1	1
Ligament ruptures	1	0	0	0	1	0
Anaesthesia	1	0	0	0	1	0
Bone maturity	1	1	0	0	0	0
Covid-19	1	0	0	0	1	0
Lumbar spinal stenosis	1	0	0	0	1	0
Musculoskeletal	1	0	0	1	0	0
Pneumonia	1	0	1	0	0	0
Possible gastric neoplasm	1	0	0	0	0	1
Skin	1	0	0	1	0	0
Varying: Pulmonary embolism, intercranial haemorrhage, and acute cervical spine fractures	1	0	1	1	0	0
Varying: Cardiac, pulmonary, and musculoskeletal	1	0	0	0	1	0
Varying: chest/lung nodules, Covid, fractures, scoliosis, prostate, neuro/dementia	1	1	0	0	0	0
Participant group						
Radiology staff		5	6	34	0	0
Wider clinical staff		1	4	4	0	0
Residents/students/trainees		0	2	10	0	0
Patients		0	0	11	0	0
Members of the public		0	0	2	0	0
Radiology staff, wider clinical staff, stakeholders, and patients		1	1	1	0	0
Radiology and wider clinical staff		0	1	0	0	0
Radiology staff and radiology students		0	0	7	0	0
Radiologists and computer scientists/industry and IT staff		0	0	3	0	0

(Table 1 continues on next page)

(Table 1 continues on next page)

Study characteristics	Overall (N = 140)	Study focus ^a				
		Implementation (N = 7)	Experiences (N = 14)	Perceptions (N = 74)	Effectiveness (quantitative) (N = 53)	Cost (N = 6)
(Continued from previous page)						
Staff (radiology and wider) and members of the public		0	0	1	0	0
Staff (radiology and wider), patients, and members of the public (including some carers)		0	0	1	0	0
Diagnostic images and readers		0	0	0	53	0
Diagnostic images		0	0	0	0	6

^aThe total will not always add up to 140 as some papers had multiple areas of focus (as shown above), were conducted across countries, covered different imaging modalities and patient pathways and recruited multiple stakeholder groups.

Table 1: Summary of study characteristics.

implementing AI in practice, with two identifying the main barriers and facilitators.^{56,58} One study referenced procurement when discussing the high cost of implementing AI,⁵⁹ but no papers explored key processes preceding implementation (e.g., procurement).

Function of AI tools

AI was used as a tool to assist with diagnosis.^{52–54} One study reported 15 different applications being used in clinical practice (8 fully integrated and 7 in exploration or implementation phases) at one medical centre, including lung nodule detection, fracture assessment, measurement, and quantification.⁵⁴

The process of implementation

The process of implementation involved integrating AI into existing radiology systems used to store and transmit images and reports.^{52,54,55} This enabled AI to run automatically in the background.⁵² One study described a holistic approach to implementing AI, where integration aligned social and technological aspects of clinical practice.⁵⁴ In this study, multiple AI algorithms were working in clinical workflows across one medical centre,⁵⁴ achieved by having one central workflow engine where data analysed by AI was sent to relevant data repositories.⁵⁴ Reasons for integrating AI into existing systems were centred around minimising workflow disruption, although some discussed teething problems (e.g., AI causing interruptions as staff adapted to viewing AI outputs).^{54,55}

Experiences of using AI

Fourteen studies explored staff and trainees' experiences of using AI (Table 1). In most studies, AI was viewed positively as a reliable and useable tool.^{33,52,59–65} AI was mainly used as a second reader (a support tool/second pair of eyes for clinicians).^{52,55,60,62,63} However, there were mixed findings about using AI in practice. In some studies, staff felt AI helped to reduce reading times,^{52,63,66} improve accuracy,^{33,63,65,66} and efficiency.^{58,65} Despite this, studies reported concerns about limited evidence,^{55,62,66} the risk of false positives,^{52,62} and reduced

efficiency.³³ In the quantitative studies evaluating effectiveness, three studies measured reader confidence level in diagnostic decisions. Findings from these studies were mixed, some found improved self-reported confidence with the use of AI⁶⁷ but others found AI did not impact confidence.^{34,67,68}

Perceptions about using AI

Seventy-four studies explored perceptions (Table 1). AI was viewed favourably in radiology^{52,55,65,69–87} as a diagnostic support tool.^{57,69,73,76,77,79,88–95} However, one paper highlighted the importance of balancing positivity towards AI in radiology with an element of scepticism.⁹⁶ Research indicated a consistent view that AI should not replace humans,^{52,55,60,65,70,80,83,84,86,89,90,97–108} demonstrating the perceived importance of continued human input. For patients, members of the public and staff, trusting AI was often reliant on human oversight,^{72,108–115} transparency, and explainability.^{58,73,114–116} Some radiology staff felt their profession is safeguarded by their autonomy,⁷⁴ as the use of AI alone is not viewed favourably⁹⁰ or legally.¹⁴ Although it was clear that AI should not replace the human element of diagnostics, studies reported that AI will likely change the radiology profession^{58,73,74,80,95,98,102,117} and there were some concerns about job security and reduced demand,^{118,119} despite increased need for medical imaging.^{1,2} Regarding how AI may change radiology, studies suggested that AI may upskill and enhance roles^{52,58} but reported a risk of reduced autonomy.⁵⁸

Factors influencing AI adoption

Table 2 provides an overview of influencing factors. Barriers included the high technical demand of AI (N = 6), lack of knowledge and training (N = 20), absence of evidence-based guidance (N = 5), complex governance processes (N = 2), and how AI lacks empathy and human connection (N = 8). Facilitators included integrating AI into existing systems (N = 4), transparency (N = 13), willingness to learn (N = 21), and having AI champions/experts (N = 8). The associated risks were staff becoming over-reliant (due to developing

	Experienced/actual (either during implementation or clinical use)	Perceived (what people think when asked about AI)
Barriers	- High technical demand and integrating AI into existing infrastructures^{52,55-58} - Absence of evidenced guidance^{53,54,56-58} - Differing expectations and knowledge^{53,54} - Uncertain and limited funding^{53,56} - Complex governance processes^{56,58}	- Lack of knowledge and training about how AI can be used in radiology specifically^{71-73,78,79,82,85-87,92-94,99,116,120-125} - Lack of organisational readiness and infrastructure to support AI^{81,88,89,101} - AI lacks human connection and empathy^{82,91,126-131}
Facilitators	- Establishing AI champions/experts who can foster a sense of leadership and openness^{53,54,56,57} - Integrating into existing systems/workflows (e.g., as staff are familiar with the systems)^{52-54,58} - Imaging and clinical implementation groups^{53,54} - Prioritising training and familiarisation^{55,58} - Openness among leaders^{53,57} - Importance of transparency^{55,57} - Willingness to learn about AI^{80,104}	- Importance of transparency—achieved by prioritising staff training^{57,101,132} and facilitating trust by explaining the use of AI^{58,109,114-116,127,129} - Expert leadership and support^{58,96,121,132} - Willingness to learn about AI^{70,71,73,77,79,80,83,85,90,95,99-102,104,105,116,120,125,133,134}
Risks	- Staff relying on AI (as a result of staff developing algorithmic bias and/or blindness)^{52,55} - AI being inaccurate/making errors¹⁰⁴ - AI causing disruption to workflow¹⁰⁴ - High costs of AI⁵⁷ - Ethical implications (e.g., uncertainties about who is responsible for AI, data privacy, safety etc.)⁵²	- Staff relying on AI (as a result of staff developing algorithmic bias and/or blindness)^{52,55,87,90,93} - AI being inaccurate/making errors^{55,60,72,82,90} - De-skilling of staff^{55,75,78,128,135} - High costs of AI^{58,72,79,87,93,95,99,118,120,133} - Ethical implications (e.g., uncertainties about who is responsible for AI, data privacy, safety etc.)^{87,88,95,103,107,108}
Benefits	- Improve accuracy^{52,65} - Improve reporting times⁵² - Reduce likelihood of follow-ups^{52,58} - Helping with routine tasks^{53,58,104} - Identifying missed diagnoses^{58,104} - Using AI for knowledge generation¹⁰⁴	- Improve accuracy^{52,60,72,75,84,87,90,91,136} - Improve reporting times^{70,90,126-128,132} - Improving workflow efficiency^{84,91,99,128,132} - Helping with routine tasks^{60,75,79,82,87,96,128}

Note: Where factors are experienced, we are referring to actual experiences of implementing or using AI in practice. Where factors are perceived, we are referring to what people think are the key influencing factors. AI – Artificial Intelligence.

Table 2: Overview of factors influencing AI adoption.

algorithmic bias and/or blindness) ($N = 5$), AI being inaccurate/making errors ($N = 6$), deskilling staff ($N = 5$), the cost of AI ($N = 11$), and the ethical implications ($N = 7$). The benefits were improving reporting times ($N = 7$), improving diagnostic accuracy/identifying missed diagnoses ($N = 12$), improving workflow efficiency ($N = 5$), and helping with routine tasks ($N = 10$).

Quantitative evidence on impact and effectiveness

Of the 53 quantitative studies, 33 measured diagnostic accuracy of AI.^{26,28,31,34,37,40,44,67,137-160} Across these studies, there was consistent evidence that AI reduces the number of incorrect positive or negative results, although some found improvements were more pronounced among less experienced staff, and any reported improvements were likely to be highly dependent on study design, implementation setting, and imaging modality.

Of the 25 studies that measured sensitivity, 19 reported improvements, 5 reported no change, and one study reported a reduction when AI was used as a diagnostic tool to assist readers (Table 3). However, some of the findings were specific to less experienced staff. Sixteen studies assessed sensitivity by reader experience level and nine reported improvements among less experienced staff only (Table 3). Of the 23 studies measuring

specificity, findings were varied, with a smaller proportion reporting improvements: 13 studies reported improvements, 7 reported no change, and 3 a reduction in specificity. Ten of the 16 studies that measured the effect of reader experience on specificity reported improvements with AI assistance (Table 3).

Evidence of improvement in time and workflow was variable, with improvements primarily seen in both reducing image interpretation time and time to report. Few studies reported on workflow impacts further along the pathway (Table 4). Although there were minor variations in definitions for interpretation time and time to report due to different study designs, interpretation time referred to time taken for a reader to interpret a single image, and time to report referred to time between image acquisition and documentation of the report. Of the 18 studies that measured image interpretation time, 13 found a reduction, while 5 reported an increase. There was also a lack of consistently observed benefits across different levels of reader experience (seniority or number of years in post) and suspicion of images (Table 4). Findings on time to report were more variable. Of the 10 studies that measured this outcome, 5 reported a reduction in reporting time, either for all images,^{27,46,47} for critical and urgent cases,¹⁴² or less experienced readers.¹⁵⁵ However, this study¹⁵⁵ also found an increase in

	Improved	No change	Reduction	Total
Sensitivity				
Total papers that measured sensitivity	19 ^{26,31,37,137-139,141-145,149,150,152,153,155,156,158,159}	5 ^{28,35,147,148,157}	1 ⁶⁷	25
Papers that assessed effect of reader experience ^a				
For all levels of experience	5 ^{31,138,141,150,156}	2 ^{147,157}	0	7
For less experienced readers only	9 ^{626,139,143-145,149,155,158,159}	0	0	9
Specificity				
Total papers that measured specificity	14 ^{31,67,137-139,141,144,149,151,153,156,158}	7 ^{28,35,145,150,155,157,159}	3 ^{26,142,148}	24
Papers that assessed effect of reader experience ^a				
For all levels of experience	4 ^{31,138,151,156}	5 ^{145,150,155,157,159}	1 ²⁶	10
For less experienced readers only	6 ^{139,141,143,144,149,158}	0	0	6

^aWe have included three studies that measured the effect of clinical specialism and reader self-efficacy here as proxies for reader experience. ^bOne paper found improvements for less experienced staff doing more difficult tasks.

Table 3: Number of studies that measured diagnostic accuracy and influence of reader experience.

reporting time for non-urgent/normal images, as these images were deprioritised despite having shorter interpretation times with the aid of AI. Four studies found no change when using AI for all images or less experienced readers (Table 4).

Six studies focused specifically on the effectiveness of AI in emergency departments (ED).^{25,28,36,41,42,158} Readers in these studies were emergency department physicians and non-specialists in radiology. Two^{28,36} found no difference in the length of stay, the rate of revisiting the ED within 30 days, nor communication times. However, one study²⁵ found shortened ED lengths of stay for patients with a confirmed diagnosis, whilst another found a small increase in median wait time to discharge.⁴¹ Two studies assessing the effect of AI on clinical decision-making found that changes to the final report, patient management, and image

recommendations were more likely for more critical images, regardless of reader experience.³⁰ One study¹⁶² found that incorrect AI results can influence a radiologist to make wrong decisions (Table 4).

Evidence on cost-effectiveness

Findings from cost-effectiveness studies ($N = 6$) were mixed. One study demonstrated that although accuracy using AI was improved, the cost-effectiveness was not, as more invasive treatment approaches generated costs over the patient's lifetime and diminished possible effectiveness advantages.³⁷ Five papers demonstrated monetary benefits and cost-effectiveness of the AI-aided pathways.^{38,48-51}

Feedback from stakeholder workshops

Both groups discussed how review findings reflected the NHS being in the early stages of using AI and learning

	Reduction	No change	Increase	Total
Interpretation time				
Total papers that measured interpretation time	13 ^{ab34,36,40,45,47,139,142,145,150,152,153,155,160}	2 ^{28,138}	5 ^{39,47,142,147,161}	18 ^a
Papers that assessed effect of reader experience				
For all levels of experience	4 ^{139,145,150,155}	1 ¹³⁸	1 ¹⁶¹	6
For <u>more</u> experienced staff only	0	0	1 ¹⁴⁷	1
Papers that assessed effect of urgency of image findings				
For non-urgent/normal images	1 ^{a142}	1 ^{b36}	0	2
For critical/urgent/more suspicious images	1 ^{b36}	0	2 ^{39,142}	3
Reporting time				
Total papers that measured reporting time	5 ^{c27,46,47,142,155}	4 ^{27,28,36,159}	2 ^{c26,142}	10 ^c
Papers that assessed effect of reader experience/priority of images				
For all levels of experience	0	0	1 ²⁶	1
For <u>less</u> experienced readers only	1 ¹⁵⁵	0	0	1
Papers that assessed effect of urgency of image findings				
For non-urgent/normal images	0	0	1 ^{c142}	1
For critical/urgent/more suspicious images	1 ^{c142}	0	0	1

^aTwo papers reported both a reduction and an increase in interpretation time depending on selected confounders, hence the reported total does not equal the sum of studies. ^bOne paper reported both a reduction and no change in interpretation time depending on selected confounders. ^cOne paper reported both a reduction and an increase in reporting time depending on selected confounders, hence the reported total does not equal the sum of studies.

Table 4: Number of studies by workflow outcome measures and findings.

through ongoing implementation. The complexity and varied implementation of AI were described as the ‘*wild west*’ by staff, with a lack of guidance and structure. However, when it comes to implementing AI in real-world settings, staff spoke on the importance of integrating AI into existing systems effectively, which causes minimal disruptions to workflow. Discussions suggested that AI would be challenging in clinical practice unless the existing context is considered. Thus, the integration of AI needs to be managed within the infrastructure of existing radiology systems.

Both groups were unsurprised about the mixed review findings in terms of experiences (e.g., some reporting AI had benefits and others reduced efficiency). The complexity of how AI is being used was noted (e.g., in different ways to achieve a range of objectives), making it difficult to provide a single answer about whether AI is beneficial. Therefore, the importance of returning to the purpose of using AI and how these technologies can help, was emphasised.

Both groups felt the review was relevant because it highlighted current evidence limitations, especially as AI is often viewed as a transformative solution. Stakeholders noted the importance of addressing evidence gaps (e.g., patient, carer and public experiences, health inequalities), and among public members, including patient voices as AI advances (example quotes in [Appendix S7](#)).

Quality appraisal of included studies

Qualitative ($N = 15$), quantitative survey, and mixed-methods studies ($N = 68$) met most of the criteria but there were some reports of sampling bias and limited description of qualitative analysis in mixed-methods studies ([Supplementary File S2](#)). The quantitative studies ($N = 53$) met most of the criteria for randomised or non-randomised studies, but many did not address potential confounders such as patient characteristics and co-morbidities. Since the findings depended on the interpretation of individual readers, the sample of readers in many studies was low, leading to risk of individual bias. Cost studies rated good ($N = 6$) ([Supplementary File S3](#)) but acknowledged factors that might impact the reliability of conclusions, including the identified range of the relevant costs, consequences for each alternative, the earliness in costs, country and facility-specific considerations and consequences identification and measurement.

Discussion

Our review highlights the paucity of research conducted in real-world settings. From the current evidence, we conclude that AI can have positive outcomes in relation to improving diagnostic accuracy and reducing interpretation times, which aligns with some early staff experiences and perceived benefits of using AI. Factors influencing implementation (e.g., high technical

demand, lack of guidance, training and knowledge, transparency, and expert engagement) were also identified. However, we do not know enough about the system-wide impact of AI, the process of procurement to implementation, experiences of using AI and/or receiving AI-based care. Current and future implementation should consider if and how AI can address the needs of healthcare systems, the implementation context and educational training needs.

The limited number of studies conducted in real-world settings aligns with research gaps highlighted in evidence generation plans.⁸ Existing research⁶ and findings from stakeholder workshops suggest this is because services are in the early stages of implementing AI,⁵⁷ with further work emerging.⁶ The positive outcomes in relation to improving diagnostic accuracy and reducing interpretation times resonate with previous literature.^{2,163,164} However, findings illustrated that improvements in diagnostic accuracy were more likely among less experienced staff² and there was evidence of AI overcalling negative findings; a risk reported in previous studies.¹⁶³ Furthermore, there was inconsistent evidence regarding experiences, how AI can improve workflow efficiency and whether these technologies are cost-effective, with few papers studying cost specifically. Variation in the quantitative findings were also likely to be dependent on study design, the imaging modality, and clinical application. From the current evidence, we cannot draw conclusions on how the potential benefits of AI may impact longer-term patient outcomes and the wider healthcare system (e.g., changes in volumes of patients for diagnostic/treatment services). Findings support previous research which shows AI has the potential to positively impact diagnostics in radiology.² In parallel, the evidence highlights caution in how AI is perceived (e.g., as a complementary tool which can help to navigate current demand, rather than a solution).¹⁶⁵

We extend previous work by reviewing literature on real-world implementation, demonstrating AI has been used to support diagnostics in a complementary role and not a replacement. This aligns with previous research,^{15,166} user guidance,¹⁴ and stakeholder views,¹⁶⁷ which highlight the central role of clinicians in maintaining human continuity (ensuring that humans have oversight and AI is not used with complete autonomy). Although AI has the potential to positively impact radiology, the synthesised literature shows that continued human oversight and transparency about how AI makes decisions, are needed to foster a sense of trust when using AI for diagnostic purposes. These findings relate to ethical concerns often associated with AI, with existing evidence recommending that AI implementation should promote safety and transparency whilst reducing risk of harm.¹⁶⁸

Furthermore, although few papers have evaluated implementation, our findings highlight the complexity of integrating AI into existing healthcare systems,

especially when organisations may not be ready to support such technical advancements,¹⁶⁶ with an absence of clear guidance.⁵⁶ Integrating AI will likely cause initial disruptions; our findings suggest that it is essential to evaluate the implementation context and ensure there is capacity to support the integration process.¹⁶⁶ Otherwise, as highlighted in our workshops, attempts to improve workflow efficiency by using AI may have the opposite effect. Variation in how AI is being used, complexity surrounding the ethical landscape and how AI can be used effectively alongside clinicians, emphasises a need for further evidence that can continue to inform clear implementation guidance and/or practice frameworks,^{58,168} such as the recently developed European Union AI Act.¹⁶⁹

We advance previous research by synthesising staff experiences of using AI. Importantly, staff and trainees had limited experience using AI; consistent with survey papers where only a small proportion of participants had used AI in a clinical setting⁵⁹ and likely reflected limited clinical implementation. No studies explored patient, carer, or public experiences of AI-based care, although workshop discussions and current policy guidance³ highlight the importance of patient voice/engagement. Therefore, the evidence needs to include the experiences of groups whose acceptance and trust are important for the ongoing use of these technologies.^{17,109}

Although evidence shows potential value in using AI, our findings suggest that implementation and use are happening ahead of developing a robust evidence base.⁸ This was reflected further in stakeholder workshops, where implementation was described as a continuous learning process rather than being evidence informed. However, AI needs to be implemented to build an evidence base that explores real-world implementation. Therefore, to ensure future use can be evidence informed, there needs to be a careful balance between implementing AI safely and conducting robust evaluations, to enable learning from important technological advancements. For evidence users, our review highlights what is already known and what needs to be considered moving forward, when interest continues to grow¹⁶⁵ and AI is used in other clinical areas.¹⁷⁰

For example, it is essential to be clear on the specific needs of healthcare systems (e.g., improving clinical outcomes and administrative efficiency), whether AI can effectively meet these needs over other solutions, and that these needs are communicated to AI developers, so that implementation is problem-driven rather than product-driven.¹⁷¹ Additionally, considerations including the healthcare pathway, country, and clinical conditions are needed, as there might be differences from setting-to-setting. Another factor is the population size as AI tools seem more effective in high-prevalence populations.³⁷ Furthermore, we highlight the need for tailored educational programmes with input from

experts, that acknowledge current knowledge and the complexity of using AI across different clinical contexts.^{2,172}

Future research should evaluate the process of implementing AI into live clinical pathways or in shadow/testing mode,⁸ including pre-implementation processes and patient, carer and public experiences of AI-based care (e.g., experiences of the diagnostic pathway featuring demonstrations,^{2,172} ethical factors like consent and transparency, patient safety, trust in AI and the impact of AI on empathy and human connection/relationships¹⁷). Such research may also evaluate how unsupervised AI (used without human supervision) is implemented, used, perceived, and experienced by different stakeholders. Although we did not include these papers, this could be an emerging focus as AI continues to be implemented globally and may be used with greater autonomy. Secondly, further research is needed on the effect of AI on patient outcomes, wider hospital systems (e.g., time to treatment, changes in volumes for other diagnostic or treatment services), diagnostic outcomes, and inequalities.⁸ Future research on cost-effectiveness of AI solutions in radiology is also needed.^{8,37} Finally, understanding the long-term impact and sustainability of AI in clinical settings is essential.³⁷

The review had a broad and inclusive focus, supported by guidance from clinical experts (FG, TOR). Findings present a summary about how AI is implemented, used, and experienced globally, as well as current evidence on effectiveness and cost, which may be relevant for healthcare systems worldwide. However, it may be difficult to generalise findings across different health systems and only papers in English were included. As AI is a rapidly evolving field, we may not have captured all evidence and papers where AI was used autonomously were out of scope. Although we searched four databases and the grey literature, not all databases were used. There is also a risk of publication bias in AI research, as in other fields. Lastly, stakeholder workshops strengthened findings by illustrating implications, but only in the context of the English NHS.

To conclude, our review suggests potential value in using AI for diagnostics in radiology, mirrored in the ongoing interest in AI. However, to assist with safe and effective procurement, implementation, validation, and evaluation, research must be planned, commissioned, and used to address the current gaps in the evidence base. This will help to draw conclusions about how best to use AI as a complementary tool in the complexities of radiology practice.

Contributors

All authors contributed to the conceptualisation and development of the study.

Data curation: RL, ED, EM, CSJ.

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Resources: RL, ED, EM, CSJ, AIGR, HW, TOR, FG, HE, RM, JL, AH.

Supervision: HW, AIGR, CSJ, SM, NJF.

Verification of the underlying data: RL, ED, EM, CSJ, HW, AIGR.

Writing – original draft: RL, ED, EM.

TOR and FG provided clinical expertise and guidance. All authors contributed to the interpretation of findings, visualisation, revising, and finalising the paper. All authors have read and approved the final manuscript.

Data sharing statement

Data will be made available upon request made to the corresponding author.

Declaration of interests

AIGR is a trustee at Health Services Research UK. RM is Chair of the Board of Trustees of the Middlesex Association for the Blind; Vice-Chair on the Board of Trustees of the Research Institute for Disabled Consumers; Trustee on the Board of Thomas Pocklington Trust; Non-Executive Director on the Board of Evenbreak; Co-chair and Director on the Board of Shaping Our Lives. SM is currently (2022-) a member of the Small Business Research Initiative (SBRI) Healthcare panel. His post is funded in part by RAND Europe, a non-profit research organisation. SM is also Deputy Director of Applied Research Collaboration East of England (NIHR ARC EoE) at Cambridgeshire and Peterborough NHS Foundation Trust. NJF was a Non-Executive Director at Whittington Health NHS Trust until October 2024, a trustee at Health Services Research UK until 2022 and is a Non-Executive Director at Covid-19 Bereaved Families for Justice UK. TOR is part of the AXREM AI Special Focus Group, the British Institute of Radiology AI Special Interest Group, NHSE AI Deployment Fund Oversight Committee and Society of Radiographers AI Advisory Group. FG is a shareholder in Optellum Ltd, is a co-founder and Chairman of the RAIQC Ltd, was an advisor to NICE on the use of chest x-ray AI in the NHS and a committee member of the RCR Advisory group. All other authors report no declarations of interest.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.eclinm.2025.103228>.

References

- Bekbolatova M, Mayer J, Ong CW, Toma M. Transformative potential of AI in healthcare: definitions, applications, and navigating the ethical landscape and public perspectives. *Healthcare (Basel)*. 2024;12(2):125. <https://doi.org/10.3390/healthcare12020125>.
- He C, Liu W, Xu J, et al. Efficiency, accuracy, and health professional's perspectives regarding artificial intelligence in radiology practice: a scoping review. *Iradiology*. 2024;2(2):156–172.
- Darzi A. Summary letter from Lord Darzi to the Secretary of State for Health and Social Care. Available from: <https://www.gov.uk/government/publications/independent-investigation-of-the-nhs-in-england/summary-letter-from-lord-darzi-to-the-secretary-of-state-for-health-and-social-care>; 2024. Accessed November 4, 2024.
- Geis JR, Brady AP, Wu CC, et al. Ethics of artificial intelligence in radiology: summary of the joint European and North American multisociety statement. *Radiology*. 2019;293(2):436–440. <https://doi.org/10.1148/radiol.2019191586>.
- Brady AP, Allen B, Chong J, et al. Developing, purchasing, implementing and monitoring AI tools in radiology: practical considerations. A multi-society statement from the ACR, CAR, ESR, RANZCR & RSNA. *J Med Imaging Radiat Oncol*. 2024;68(1):7–26. <https://doi.org/10.1111/1754-9485.13612>.
- Gleeson F, Revel MP, Biederer J, et al. Implementation of artificial intelligence in thoracic imaging-a what, how, and why guide from the European Society of Thoracic Imaging (ESTI). *Eur Radiol*. 2023;33(7):5077–5086. <https://doi.org/10.1007/s00330-023-09409-2>.
- van Leeuwen KG, de Rooij M, Schalekamp S, van Ginneken B, Rutten MJCM. How does artificial intelligence in radiology improve efficiency and health outcomes? *Pediatr Radiol*. 2022;52(11):2087–2093. <https://doi.org/10.1007/s00247-021-05114-8>.
- National Institute for Health and Care Excellence (NICE). Evidence generation plan. Available from: <https://www.nice.org.uk/consultations/2576/4/evidence-gaps>; 2024. Accessed September 10, 2024.
- Rapid Service Evaluation Team. Using artificial intelligence in chest diagnostics for lung disease. Available from: <https://www.nuffieldtrust.org.uk/rset-rapid-evaluations-of-new-ways-of-providing-care/projects/using-artificial-intelligence-in-chest-diagnostics-for-lung-disease>; 2024. Accessed November 4, 2024.
- Tricco CA, Langlois VE, Straus ES. *Rapid Reviews to Strengthen Health Policy Systems: A Practical Guide*. Geneva: World Health Organization; 2017.
- Moher D, Liberati A, Tetzlaff J, Altman DG, PRISMA Group. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med*. 2009;6(7):e1000097. <https://doi.org/10.1371/journal.pmed.1000097>.
- Peters MD, Godfrey CM, Khalil H, McInerney P, Parker D, Soares CB. Guidance for conducting systematic scoping reviews. *Int J Evid Based Healthc*. 2015;13(3):141–146. <https://doi.org/10.1097/XEB.0000000000000050>.
- Garrity C, Gartlehner G, Nussbaumer-Streit B, et al. Cochrane Rapid Reviews Methods Group offers evidence-informed guidance to conduct rapid reviews. *J Clin Epidemiol*. 2021;130:13–22. <https://doi.org/10.1016/j.jclinepi.2020.10.007>.
- UK Statutory Instruments. The ionising radiation (medical exposure) (amendment) regulations 2024. Available from: <https://www.legislation.gov.uk/ukxi/2024/896/made>; 2024. Accessed October 7, 2024.
- National Institute for Health and Care Excellence (NICE). Artificial intelligence-derived software to analyse chest X-rays for suspected lung cancer in primary care referrals: early value assessment. Available from: <https://www.nice.org.uk/guidance/hte12>; 2023. Accessed September 2, 2024.
- Hogg HDJ, Al-Zubaidy M, Technology Enhanced Macular Services Study Reference Group, et al. Stakeholder perspectives of clinical artificial intelligence implementation: systematic review of qualitative evidence. *J Med Internet Res*. 2023;25:e39742. <https://doi.org/10.2196/39742>.
- Young AT, Amara D, Bhattacharya A, Wei ML. Patient and general public attitudes towards clinical artificial intelligence: a mixed methods systematic review. *Lancet Digit Health*. 2021;3(9):e599–e611. [https://doi.org/10.1016/S2589-7500\(21\)00132-1](https://doi.org/10.1016/S2589-7500(21)00132-1).
- The EndNote Team. *EndNote*. Philadelphia, PA: Clarivate; 2013.
- Ouzzani M, Hammady H, Fedorowicz Z, Elmagarmid A. Rayyan — a web and mobile app for systematic reviews. *Systematic Reviews*. 2016;5(1):210. <https://doi.org/10.1186/s13643-016-0384-4>.
- Popay J, Roberts H, Sowden A, et al. *Guidance on the Conduct of Narrative Synthesis in Systematic Reviews*. Lancaster: Lancaster University; 2006.
- Hong QN, Pluye P, Fàbregues S, et al. Improving the content validity of the mixed methods appraisal tool: a modified e-Delphi study. *J Clin Epidemiol*. 2019;111:49–59.e1. <https://doi.org/10.1016/j.jclinepi.2019.03.008>.
- Drummond MF, Sculpher MJ, Claxton K, Stoddart GL, Torrance GW. *Methods for the Economic Evaluation of Health Care Programmes*. Oxford: Oxford University Press; 2015.
- Doran CM. Economic evaluation of interventions to treat opiate dependence: a review of the evidence. *Pharmacoeconomics*. 2008;26(5):371–393. <https://doi.org/10.2165/00019053-200826050-00003>.

- 24 Braun V, Clarke V. Using thematic analysis in psychology. *Qual Res Psychol*. 2006;3(2):77–101.
- 25 Chien HC, Yang TL, Juang WC, Chen YA, Li YJ, Chen CY. Pilot report for intracranial hemorrhage detection with deep learning implanted head computed tomography images at emergency department. *J Med Syst*. 2022;46(7):49. <https://doi.org/10.1007/s10916-022-01833-z>.
- 26 Forookhi A, Laschena L, Pecoraro M, et al. Bridging the experience gap in prostate multiparametric magnetic resonance imaging using artificial intelligence: a prospective multi-reader comparison study on inter-reader agreement in PI-RADS v2.1, image quality and reporting time between novice and expert readers. *Eur J Radiol*. 2023;161:110749. <https://doi.org/10.1016/j.ejrad.2023.110749>.
- 27 Hong W, Hwang EJ, Park CM, Goo JM. Effects of implementing artificial intelligence-based computer-aided detection for chest radiographs in daily practice on the rate of referral to chest computed tomography in pulmonology outpatient clinic. *Korean J Radiol*. 2023;24(9):890–902. <https://doi.org/10.3348/kjr.2023.0255>.
- 28 Hwang EJ, Goo JM, Nam JG, Park CM, Hong KJ, Kim KH. Conventional versus artificial intelligence-assisted interpretation of chest radiographs in patients with acute respiratory symptoms in emergency department: a pragmatic randomized clinical trial. *Korean J Radiol*. 2023;24(3):259–270. <https://doi.org/10.3348/kjr.2022.0651>.
- 29 Ishiyama M, Kudo SE, Misawa M, et al. Impact of the clinical use of artificial intelligence-assisted neoplasia detection for colonoscopy: a large-scale prospective, propensity score-matched study (with video). *Gastrointest Endosc*. 2022;95(1):155–163. <https://doi.org/10.1016/j.gie.2021.07.022>.
- 30 Jones CM, Danaher L, Milne MR, et al. Assessment of the effect of a comprehensive chest radiograph deep learning model on radiologist reports and patient outcomes: a real-world observational study. *BMJ Open*. 2021;11(12):e052902. <https://doi.org/10.1136/bmjopen-2021-052902>.
- 31 Li J, Zhou L, Zhan Y, et al. How does the artificial intelligence-based image-assisted technique help physicians in diagnosis of pulmonary adenocarcinoma? A randomized controlled experiment of multicenter physicians in China. *J Am Med Inform Assoc*. 2022;29(12):2041–2049. <https://doi.org/10.1093/jamia/ocac179>.
- 32 Liu P, Wang P, Glissen Brown JR, et al. The single-monitor trial: an embedded CADE system increased adenoma detection during colonoscopy: a prospective randomized study. *Therap Adv Gastroenterol*. 2020;13:1756284820979165. <https://doi.org/10.1177/1756284820979165>.
- 33 Nehme F, Coronel E, Barringer DA, et al. Performance and attitudes toward real-time computer-aided polyp detection during colonoscopy in a large tertiary referral center in the United States. *Gastrointest Endosc*. 2023;98(1):100–109.e6. <https://doi.org/10.1016/j.gie.2023.02.016>.
- 34 Nhat PTH, Van Hao N, Tho PV, et al. Clinical benefit of AI-assisted lung ultrasound in a resource-limited intensive care unit. *Crit Care*. 2023;27(1):257. <https://doi.org/10.1186/s13054-023-04548-w>.
- 35 Rosa F, Buccicardi D, Romano A, Borda F, D'Auria MC, Gastaldo A. Artificial intelligence and pelvic fracture diagnosis on X-rays: a preliminary study on performance, workflow integration and radiologists' feedback assessment in a spoke emergency hospital. *Eur J Radiol Open*. 2023;11:100504. <https://doi.org/10.1016/j.ejro.2023.100504>.
- 36 Schmuelling L, Franzeck FC, Nickel CH, et al. Deep learning-based automated detection of pulmonary embolism on CT pulmonary angiograms: no significant effects on report communication times and patient turnaround in the emergency department nine months after technical implementation. *Eur J Radiol*. 2021;141:109816. <https://doi.org/10.1016/j.ejrad.2021.109816>.
- 37 Schwendicke F, Rossi JG, Göstemeyer G, et al. Cost-effectiveness of artificial intelligence for proximal caries detection. *J Dent Res*. 2021;100(4):369–376. <https://doi.org/10.1177/0022034520972335>.
- 38 Thiruvengadam NR, Solaimani P, Shrestha M, et al. The efficacy of real-time computer-aided detection of colonic neoplasia in community practice: a pragmatic randomized controlled trial. *Clin Gastroenterol Hepatol*. 2024;22(11):2221–2230.e15. <https://doi.org/10.1016/j.cgh.2024.02.021>.
- 39 Wenderott K, Krups J, Luetkens JA, Gambashidze N, Weigl M. Prospective effects of an artificial intelligence-based computer-aided detection system for prostate imaging on routine workflow and radiologists' outcomes. *Eur J Radiol*. 2024;170:111252. <https://doi.org/10.1016/j.ejrad.2023.111252>.
- 40 Yacoub B, Varga-Szemes A, Schoepf UJ, et al. Impact of artificial intelligence assistance on chest CT interpretation times: a prospective randomized study. *AJR Am J Roentgenol*. 2022;219(5):743–751. <https://doi.org/10.2214/AJR.22.27598>.
- 41 Chiramal JA, Johnson J, Webster J, et al. Artificial Intelligence-based automated CT brain interpretation to accelerate treatment for acute stroke in rural India: an interrupted time series study. *PLOS Glob Public Health*. 2024;4(7):e0003351. <https://doi.org/10.1371/journal.pgph.0003351>.
- 42 Herpe G, Nelken H, Vendevre T, et al. Effectiveness of an artificial intelligence software for limb radiographic fracture recognition in an emergency department. *J Clin Med*. 2024;13(18):5575. <https://doi.org/10.3390/jcm13185575>.
- 43 Glessgen C, Crowe LA, Wetzl J, et al. Automated vs manual cardiac MRI planning: a single-center prospective evaluation of reliability and scan times. *Eur Radiol*. 2025. <https://doi.org/10.1007/s00330-025-11364-z>.
- 44 Deng R, Liu Y, Wang K, et al. Comparison of MRI artificial intelligence-guided cognitive fusion-targeted biopsy versus routine cognitive fusion-targeted prostate biopsy in prostate cancer diagnosis: a randomized controlled trial. *BMC Med*. 2024;22(1):530. <https://doi.org/10.1186/s12916-024-03742-z>.
- 45 Del Gaizo AJ, Osborne TF, Shahoumian T, Sherrier R. Deep learning to detect intracranial hemorrhage in a national tele-radiology program and the impact on interpretation time. *Radiol Artif Intell*. 2024;6(5):e240067. <https://doi.org/10.1148/ryai.240067>.
- 46 Hunter JG, Bera K, Shah N, et al. Real-world performance of pneumothorax-detecting artificial intelligence algorithm and its impact on radiologist reporting times. *Acad Radiol*. 2024;32(3):1165–1174. <https://doi.org/10.1016/j.acra.2024.10.012>.
- 47 Savage CH, Elkassem AA, Hamki O, et al. Prospective evaluation of artificial intelligence triage of incidental pulmonary emboli on contrast-enhanced CT examinations of the chest or abdomen. *AJR Am J Roentgenol*. 2024;223(3):e2431067. <https://doi.org/10.2214/ajr.24.31067>.
- 48 van Leeuwen KG, Meijer FJA, Schalekamp S, et al. Cost-effectiveness of artificial intelligence aided vessel occlusion detection in acute stroke: an early health technology assessment. *Insights Imaging*. 2021;12(1):133. <https://doi.org/10.1186/s13244-021-01077-4>.
- 49 Thiruvengadam NR, Coté GA, Gupta S, et al. An evaluation of critical factors for the cost-effectiveness of real-time computer-aided detection: sensitivity and threshold analyses using a micro-simulation model. *Gastroenterology*. 2023;164(6):906–920. <https://doi.org/10.1053/j.gastro.2023.01.027>.
- 50 Yonazu S, Ozawa T, Nakanishi T, et al. Cost-effectiveness analysis of the artificial intelligence diagnosis support system for early gastric cancers. *DEN Open*. 2024;4(1):e289. <https://doi.org/10.1002/deo2.289>.
- 51 Brix MAK, Järvinen J, Bode MK, et al. Financial impact of incorporating deep learning reconstruction into magnetic resonance imaging routine. *Eur J Radiol*. 2024;175:111434. <https://doi.org/10.1016/j.ejrad.2024.111434>.
- 52 Faric N, Hinder S, Williams R, et al. Early experiences of integrating an artificial intelligence-based diagnostic decision support system into radiology settings: a qualitative study. *Stud Health Technol Inform*. 2023;309:240–241. <https://doi.org/10.3233/SHTI230787>.
- 53 Strohm L, Hehakaya C, Ranschaert ER, Boon WPC, Moors EHM. Implementation of artificial intelligence (AI) applications in radiology: hindering and facilitating factors. *Eur Radiol*. 2020;30(10):5525–5532. <https://doi.org/10.1007/s00330-020-06946-y>.
- 54 Kim B, Romeijn S, van Buchem M, Mehrizi MHR, Grootjans W. A holistic approach to implementing artificial intelligence in radiology. *Insights Imaging*. 2024;15(1):22. <https://doi.org/10.1186/s13244-023-01586-4>.
- 55 Wenderott K, Krups J, Luetkens JA, Weigl M. Radiologists' perspectives on the workflow integration of an artificial intelligence-based computer-aided detection system: a qualitative study. *Appl Ergon*. 2024;117:104243. <https://doi.org/10.1016/j.apergo.2024.104243>.
- 56 Royal College of Radiologists. *Overcoming Barriers to AI Implementation in Imaging: Outcomes of an RCR Expert Stakeholder Day*. London: The Royal College of Radiologists; 2023.
- 57 NHS England. *Understanding Healthcare Workers' Confidence in AI*. England: NHS England; 2022.
- 58 Stogiannos N, O'Regan T, Scurr E, et al. Lessons on AI implementation from senior clinical practitioners: an exploratory qualitative study in medical imaging and radiotherapy in the UK. *J Med Imaging Radiat Sci*. 2025;56(1):101797. <https://doi.org/10.1016/j.jmir.2024.101797>.

- 59 European Society of Radiology (ESR).. Current practical experience with artificial intelligence in clinical radiology: a survey of the European Society of Radiology. *Insights Imaging*. 2022;13(1):107. <https://doi.org/10.1186/s13244-022-01247-y>.
- 60 Choi H, Sunwoo L, Cho SJ, et al. A nationwide web-based survey of neuroradiologists' perceptions of artificial intelligence software for neuro-applications in Korea. *Korean J Radiol*. 2023;24(5):454–464. <https://doi.org/10.3348/kjr.2022.0905>.
- 61 Carlile M, Hurt B, Hsiao A, Hogarth M, Longhurst CA, Dameff C. Deployment of artificial intelligence for radiographic diagnosis of COVID-19 pneumonia in the emergency department. *J Am Coll Emerg Physicians Open*. 2020;1(6):1459–1464. <https://doi.org/10.1002/emp2.12297>.
- 62 Rabinovich D, Mosquera C, Torrens P, Aineseder M, Benitez S. User satisfaction with an AI system for chest X-ray analysis implemented in a hospital's emergency setting. *Stud Health Technol Inform*. 2022;294:8–12. <https://doi.org/10.3233/SHIT220386>.
- 63 Shin HJ, Lee S, Kim S, Son NH, Kim EK. Hospital-wide survey of clinical experience with artificial intelligence applied to daily chest radiographs. *PLoS One*. 2023;18(3):e0282123. <https://doi.org/10.1371/journal.pone.0282123>.
- 64 Yoon DH, Heo S, Yu JY, et al. Effect of an artificial-intelligent chest radiographs reporting system in an emergency department. *Signa Vitae*. 2023;19(6):144–151. <https://doi.org/10.22514/sv.2023.108>.
- 65 Liu W, Wu Y, Zheng Z, Yu W, Bittle MJ, Kharrazi H. Evaluating artificial intelligence's role in lung nodule diagnostics: a survey of radiologists in two pilot tertiary hospitals in China. *J Clin Imaging Sci*. 2024;14:31. https://doi.org/10.25259/jcis_72_2024.
- 66 Hoppe BF, Rueckel J, Dikhtyar Y, et al. Implementing artificial intelligence for emergency radiology impacts physicians' knowledge and perception: a prospective pre- and post-analysis. *Invest Radiol*. 2024;59(5):404–412. <https://doi.org/10.1097/RLI.0000000000001034>.
- 67 Meng F, Kottlors J, Shahzad R, et al. AI support for accurate and fast radiological diagnosis of COVID-19: an international multicenter, multivendor CT study. *Eur Radiol*. 2023;33(6):4280–4291. <https://doi.org/10.1007/s00330-022-09335-9>.
- 68 Bowness JS, Macfarlane AJR, Burckett-St Laurent D, et al. Evaluation of the impact of assistive artificial intelligence on ultrasound scanning for regional anaesthesia. *Br J Anaesth*. 2023;130(2):226–233. <https://doi.org/10.1016/j.bja.2022.07.049>.
- 69 Agrawal A, Khatri GD, Khurana B, Sodickson AD, Liang Y, Dreizin D. A survey of ASER members on artificial intelligence in emergency radiology: trends, perceptions, and expectations. *Emerg Radiol*. 2023;30(3):267–277. <https://doi.org/10.1007/s10140-023-02121-0>.
- 70 Akinmoladun JA, Smart AE, Atalabi OM. Knowledge, attitude, and perception of radiologists about artificial intelligence in Nigeria. *W Afr J Radiol*. 2022;29(2):112–117. https://doi.org/10.4103/wajr.wajr_42_21.
- 71 Al Mohammad B, Aldaradkeh A, Gharaibeh M, Reed W. Assessing radiologists' and radiographers' perceptions on artificial intelligence integration: opportunities and challenges. *Br J Radiol*. 2024;97(1156):763–769. <https://doi.org/10.1093/bjr/tqae022>.
- 72 Aldhafeeri FM. Perspectives of radiographers on the emergence of artificial intelligence in diagnostic imaging in Saudi Arabia. *Insights Imaging*. 2022;13(1):178. <https://doi.org/10.1186/s13244-022-01319-z>.
- 73 Botwe BO, Akudjedu TN, Antwi WK, et al. The integration of artificial intelligence in medical imaging practice: perspectives of African radiographers. *Radiography (Lond)*. 2021;27(3):861–866. <https://doi.org/10.1016/j.radi.2021.01.008>.
- 74 Chen Y, Stavropoulou C, Narasinkam R, Baker A, Scarbrough H. Professionals' responses to the introduction of AI innovations in radiology and their implications for future adoption: a qualitative study. *BMC Health Serv Res*. 2021;21(1):813. <https://doi.org/10.1186/s12913-021-06861-y>.
- 75 Coppola F, Faggioni L, Regge D, et al. Artificial intelligence: radiologists' expectations and opinions gleaned from a nationwide online survey. *Radiol Med*. 2021;126(1):63–71. <https://doi.org/10.1007/s11547-020-01205-y>.
- 76 Eschert T, Schwendicke F, Krois J, Bohner L, Vinayahalingam S, Hanisch M. A survey on the use of artificial intelligence by clinicians in dentistry and oral and maxillofacial surgery. *Medicina (Kaunas)*. 2022;58(8):5. <https://doi.org/10.3390/medicina58081059>.
- 77 Pedersen MRV, Kusk MW, Lysdahlgaard S, Mork-Knudsen H, Malamateniou C, Jensen J. Nordic radiographers' and students' perspectives on artificial intelligence - a cross-sectional online survey. *Radiography (Lond)*. 2024;30(3):776–783. <https://doi.org/10.1016/j.radi.2024.02.020>.
- 78 Ryan ML, O'Donovan T, McNulty JP. Artificial intelligence: the opinions of radiographers and radiation therapists in Ireland. *Radiography (Lond)*. 2021;27(Suppl 1):S74–S82. <https://doi.org/10.1016/j.radi.2021.07.022>.
- 79 Scheetz J, Rothschild P, McGuinness M, et al. A survey of clinicians on the use of artificial intelligence in ophthalmology, dermatology, radiology and radiation oncology. *Sci Rep*. 2021;11(1):5193. <https://doi.org/10.1038/s41598-021-84698-5>.
- 80 Sharip H, Che Zakaria WFW, Leong SS, Ali Masoud M, Mohd Junaidi MZH. Radiographers' acceptance on the integration of artificial intelligence into medical imaging practice. Kuala Terengganu, Malaysia. *Environ Behav Proc J*. 2023;8(25):255–260. <https://doi.org/10.21834/e-bpj.v8i25.4872>.
- 81 Hardie T, Horton T, Willis M, Warburton W. *Switched on: How Do We Get The Best Out of Automation and AI in Health Care?* United Kingdom (UK): The Health Foundation; 2021.
- 82 The Health Foundation. AI in health care: what do the public and NHS staff think?. Available from: <https://www.health.org.uk/publications/long-reads/ai-in-health-care-what-do-the-public-and-nhs-staff-think>; 2024. Accessed May 9, 2024.
- 83 Adelsmayr G, Janisch M, Pohl M, Fuchsjäger M, Schöllnast H. Facing the AI challenge in radiology: lessons learned from a regional survey among Austrian radiologists in academic and non-academic settings on perceptions and expectations towards artificial intelligence. *Digit Health*. 2024;10:20552076241298472. <https://doi.org/10.1177/20552076241298472>.
- 84 Cè M, Ibba S, Cellina M, et al. Radiologists' perceptions on AI integration: an in-depth survey study. *Eur J Radiol*. 2024;177:111590. <https://doi.org/10.1016/j.ejrad.2024.111590>.
- 85 Goyal S, Sakhi P, Kalidindi S, Nema D, Pakhare AP. Knowledge, attitudes, perceptions, and practices related to artificial intelligence in radiology among Indian radiologists and residents: a multicenter nationwide study. *Cureus*. 2024;16(12):e76667. <https://doi.org/10.7759/cureus.76667>.
- 86 Huang W, Li Y, Bao Z, et al. Knowledge, attitude and practice of radiologists regarding artificial intelligence in medical imaging. *J Multidiscip Healthc*. 2024;17:3109–3119. <https://doi.org/10.2147/jmdh.S451301>.
- 87 Pedersen MRV, Kusk MW, Lysdahlgaard S, Mork-Knudsen H, Malamateniou C, Jensen J. A Nordic survey on artificial intelligence in the radiography profession - is the profession ready for a culture change? *Radiography (Lond)*. 2024;30(4):1106–1115. <https://doi.org/10.1016/j.radi.2024.04.020>.
- 88 Currie G, Nelson T, Hewis J, et al. Australian perspectives on artificial intelligence in medical imaging. *J Med Radiat Sci*. 2022;69(3):282–292. <https://doi.org/10.1002/jmrs.581>.
- 89 Huisman M, Ranschaert E, Parker W, et al. An international survey on AI in radiology in 1041 radiologists and radiology residents part 2: expectations, hurdles to implementation, and education. *Eur Radiol*. 2021;31(11):8797–8806. <https://doi.org/10.1007/s00330-021-07782-4>.
- 90 Lombi L, Rossero E. How artificial intelligence is reshaping the autonomy and boundary work of radiologists. A qualitative study. *Sociol Health Illn*. 2024;46(2):200–218. <https://doi.org/10.1111/1467-9566.13702>.
- 91 Ng CT, Roslan SNA, Chng YH, et al. Singapore radiographers' perceptions and expectations of artificial intelligence - a qualitative study. *J Med Imaging Radiat Sci*. 2022;53(4):554–563. <https://doi.org/10.1016/j.jmir.2022.08.005>.
- 92 Rainey C, O'Regan T, Matthew J, et al. Beauty is in the AI of the beholder: are we ready for the clinical integration of artificial intelligence in radiography? An exploratory analysis of perceived AI knowledge, skills, confidence, and education perspectives of UK radiographers. *Front Digit Health*. 2021;3:739327. <https://doi.org/10.3389/fdgh.2021.739327>.
- 93 Wadhwa V, Alagappan M, Gonzalez A, et al. Physician sentiment toward artificial intelligence (AI) in colonoscopic practice: a survey of US gastroenterologists. *Endosc Int Open*. 2020;8(10):E1379–E1384. <https://doi.org/10.1055/a-1223-1926>.
- 94 Abuzaid MM, Elshami W, Tekin H, Issa B. Assessment of the willingness of radiologists and radiographers to accept the integration of artificial intelligence into radiology practice. *Acad Radiol*. 2022;29(1):87–94. <https://doi.org/10.1016/j.acra.2020.09.014>.
- 95 Arif WM. Radiologic technology students' perceptions on adoption of artificial intelligence technology in radiology. *Int J Gen Med*. 2024;17:3129–3136. <https://doi.org/10.2147/ijgm.S465944>.
- 96 Drogot J, Milota M, Veldhuis W, Vos S, Jongsma K. The promise of AI for image-driven medicine: qualitative interview study of

- radiologists' and pathologists' perspectives. *JMIR Hum Factors*. 2024;11:e52514. <https://doi.org/10.2196/52514>.
- 97 Abuzaid MM, Elshami W, McConnell J, Tekin HO. An extensive survey of radiographers from the Middle East and India on artificial intelligence integration in radiology practice. *Health Technol (Berl)*. 2021;11(5):1045–1050. <https://doi.org/10.1007/s12553-021-00583-1>.
 - 98 Meshari Ali AA, Ahmad Alhadlaq F, Saleh Alabdullatif G, et al. Medical student's attitudes and perceptions toward artificial intelligence applications. *J Educ Teach Train*. 2022;13(4):151–157. <https://doi.org/10.47750/jett.2022.13.04.021>.
 - 99 Alsharif W, Qurashi A, Toonsi F, et al. A qualitative study to explore opinions of Saudi Arabian radiologists concerning AI-based applications and their impact on the future of the radiology. *BJR Open*. 2022;4(1):20210029. <https://doi.org/10.1259/bjro.20210029>.
 - 100 Coakley S, Young R, Moore N, et al. Radiographers' knowledge, attitudes and expectations of artificial intelligence in medical imaging. *Radiography (Lond)*. 2022;28(4):943–948. <https://doi.org/10.1016/j.radi.2022.06.020>.
 - 101 Edzie EKM, Dzefi-Tettey K, Asemah AR, et al. Perspectives of radiologists in Ghana about the emerging role of artificial intelligence in radiology. *Heliyon*. 2023;9(5):e15558. <https://doi.org/10.1016/j.heliyon.2023.e15558>.
 - 102 Eltorai AEM, Bratt AK, Guo HH. Thoracic radiologists' versus computer scientists' perspectives on the future of artificial intelligence in radiology. *J Thorac Imaging*. 2020;35(4):255–259. <https://doi.org/10.1097/RTI.0000000000000453>.
 - 103 Jungmann F, Jorg T, Hahn F, et al. Attitudes toward artificial intelligence among radiologists, IT specialists, and industry. *Acad Radiol*. 2021;28(6):834–840. <https://doi.org/10.1016/j.acra.2020.04.011>.
 - 104 Shiang T, Garwood E, Debenedictis CM. Artificial intelligence-based decision support system (AI-DSS) implementation in radiology residency: introducing residents to AI in the clinical setting. *Clin Imaging*. 2022;92:32–37. <https://doi.org/10.1016/j.clinimag.2022.09.003>.
 - 105 Rainey C, O'Regan T, Matthew J, et al. UK reporting radiographers' perceptions of AI in radiographic image interpretation - current perspectives and future developments. *Radiography (Lond)*. 2022;28(4):881–888. <https://doi.org/10.1016/j.radi.2022.06.006>.
 - 106 Lim SS, Phan TD, Law M, et al. Non-radiologist perception of the use of artificial intelligence (AI) in diagnostic medical imaging reports. *J Med Imaging Radiat Oncol*. 2022;66(8):1029–1034. <https://doi.org/10.1111/1754-9485.13388>.
 - 107 Alelyani M, Alamri S, Alqahtani MS, et al. Radiology community attitude in Saudi Arabia about the applications of artificial intelligence in radiology. *Healthcare (Basel)*. 2021;9(7):834. <https://doi.org/10.3390/healthcare9070834>.
 - 108 Fransen SJ, Kwee TC, Rouw D, et al. Patient perspectives on the use of artificial intelligence in prostate cancer diagnosis on MRI. *Eur Radiol*. 2025;35(2):769–775. <https://doi.org/10.1007/s00330-024-11012-y>.
 - 109 Clements W, Thong LP, Zia A, Moriarty HK, Goh GS. A prospective study assessing patient perception of the use of artificial intelligence in radiology. *Asia Pac J Health Manag*. 2022;17(1):46–55. <https://doi.org/10.24083/apjhm.v17i1.861>.
 - 110 Ibbas S, Tancredi C, Fantesini A, et al. How do patients perceive the AI-radiologists interaction? Results of a survey on 2119 responders. *Eur J Radiol*. 2023;165:110917. <https://doi.org/10.1016/j.ejrad.2023.110917>.
 - 111 Lennartz S, Dratsch T, Zopf D, et al. Use and control of artificial intelligence in patients across the medical workflow: single-center questionnaire study of patient perspectives. *J Med Internet Res*. 2021;23(2):e24221. <https://doi.org/10.2196/24221>.
 - 112 Rodler S, Kopliuk R, Ulrich D, et al. Patients' trust in artificial intelligence-based decision-making for localized prostate cancer: results from a prospective trial. *Eur Urol Focus*. 2024;10(4):654–661. <https://doi.org/10.1016/j.euf.2023.10.020>.
 - 113 York T, Jenney H, Jones G. Clinician and computer: a study on patient perceptions of artificial intelligence in skeletal radiography. *BMJ Health Care Inform*. 2020;27(3):e100233. <https://doi.org/10.1136/bmjhci-2020-100233>.
 - 114 Baghdadi LR, Mobeirek AA, Alhudaithi DR, et al. Patients' attitudes toward the use of artificial intelligence as a diagnostic tool in radiology in Saudi Arabia: cross-sectional study. *JMIR Hum Factors*. 2024;11:e53108. <https://doi.org/10.2196/53108>.
 - 115 Xuereb F, Portelli DJL. The knowledge and perception of patients in Malta towards artificial intelligence in medical imaging. *J Med Imaging Radiat Sci*. 2024;55(4):101743. <https://doi.org/10.1016/j.jmir.2024.101743>.
 - 116 Aldhafeeri FM. Navigating the ethical landscape of artificial intelligence in radiography: a cross-sectional study of radiographers' perspectives. *BMC Med Ethics*. 2024;25(1):52. <https://doi.org/10.1186/s12910-024-01052-w>.
 - 117 Rainey C, O'Regan T, Matthew J, et al. An insight into the current perceptions of UK radiographers on the future impact of AI on the profession: a cross-sectional survey. *J Med Imaging Radiat Sci*. 2022;53(3):347–361. <https://doi.org/10.1016/j.jmir.2022.05.010>.
 - 118 Antwi WK, Akudjedu TN, Botwe BO. Artificial intelligence in medical imaging practice in Africa: a qualitative content analysis study of radiographers' perspectives. *Insights Imaging*. 2021;12(1):80. <https://doi.org/10.1186/s13244-021-01028-z>.
 - 119 Chen Y, Wu Z, Wang P, et al. Radiology residents' perceptions of artificial intelligence: nationwide cross-sectional survey study. *J Med Internet Res*. 2023;25:e48249. <https://doi.org/10.2196/48249>.
 - 120 Huisman M, Ranschaert E, Parker W, et al. An international survey on AI in radiology in 1,041 radiologists and radiology residents part 1: fear of replacement, knowledge, and attitude. *Eur Radiol*. 2021;31(9):7058–7066. <https://doi.org/10.1007/s00330-021-07781-5>.
 - 121 Stogiannos N, O'Regan T, Scurr E, et al. AI implementation in the UK landscape: knowledge of AI governance, perceived challenges and opportunities, and ways forward for radiographers. *Radiography (Lond)*. 2024;30(2):612–621. <https://doi.org/10.1016/j.radi.2024.01.019>.
 - 122 Allam AH, Eltewacy NK, Alabdallat YJ, et al. Knowledge, attitude, and perception of Arab medical students towards artificial intelligence in medicine and radiology: a multi-national cross-sectional study. *Eur Radiol*. 2024;34(7):1–14. <https://doi.org/10.1007/s00330-023-10509-2>.
 - 123 Caparros Galan G, Sendra Portero F. Medical students' perceptions of the impact of artificial intelligence in radiology. *Radiologia (Engl Ed)*. 2022;64(6):516–524. <https://doi.org/10.1016/j.rxeng.2021.03.008>.
 - 124 Qurashi AA, Alanazi RK, Alhazmi YM, Almohammadi AS, Alsharif WM, Alshamrani KM. Saudi radiology personnel's perceptions of artificial intelligence implementation: a cross-sectional study. *J Multidiscip Healthc*. 2021;14:3225–3231. <https://doi.org/10.2147/JMDH.S340786>.
 - 125 Hamd ZY, Alorainy AI, Aldhahi MI, et al. Evaluation of the impact of artificial intelligence on clinical practice of radiology in Saudi Arabia. *J Multidiscip Healthc*. 2024;17:4745–4756. <https://doi.org/10.2147/jmdh.S465508>.
 - 126 Adams SJ, Tang R, Babyn P. Patient perspectives and priorities regarding artificial intelligence in radiology: opportunities for patient-centered radiology. *J Am Coll Radiol*. 2020;17(8):1034–1036. <https://doi.org/10.1016/j.jacr.2020.01.007>.
 - 127 Bahakeem BH, Alobaidi SF, Alzahrani AS, et al. The general population's perspectives on implementation of artificial intelligence in radiology in the western region of Saudi Arabia. *Cureus*. 2023;15(4):e37391. <https://doi.org/10.7759/cureus.37391>.
 - 128 Akudjedu TN, Torre S, Khine R, Katsifarakis D, Newman D, Malamateniou C. Knowledge, perceptions, and expectations of Artificial intelligence in radiography practice: a global radiography workforce survey. *J Med Imaging Radiat Sci*. 2023;54(1):104–116. <https://doi.org/10.1016/j.jmir.2022.11.016>.
 - 129 Zhang Z, Citardi D, Wang D, Genc Y, Shan J, Fan X. Patients' perceptions of using artificial intelligence (AI)-based technology to comprehend radiology imaging data. *Health Informatics J*. 2021;27(2):14604582211011215. <https://doi.org/10.1177/14604582211011215>.
 - 130 Jutzi TB, Kriehoff-Henning EI, Holland-Letz T, et al. Artificial intelligence in skin cancer diagnostics: the patients' perspective. *Front Med (Lausanne)*. 2020;7:233. <https://doi.org/10.3389/fmed.2020.00233>.
 - 131 Miro Catalina Q, Femenia J, Fuster-Casanovas A, et al. Knowledge and perception of the use of AI and its implementation in the field of radiology: cross-sectional study. *J Med Internet Res*. 2023;25:e50728. <https://doi.org/10.2196/50728>.
 - 132 Stogiannos N, Litosseliti L, O'Regan T, et al. Black box no more: a cross-sectional multi-disciplinary survey for exploring governance and guiding adoption of AI in medical imaging and radiotherapy in the UK. *Int J Med Inform*. 2024;186:105423. <https://doi.org/10.1016/j.ijmedinf.2024.105423>.
 - 133 Barreiro-Ares A, Morales-Santiago A, Sendra-Portero F, Souto-Bayarri M. Impact of the rise of artificial intelligence in radiology: what do students think? *Int J Environ Res Public Health*. 2023;20(2):1589. <https://doi.org/10.3390/ijerph20021589>.
 - 134 Angkurawaranon S, Inmuto N, Bannangkoon K, et al. Attitudes and perceptions of Thai medical students regarding artificial intelligence in radiology and medicine. *BMC Med Educ*. 2024;24(1):1188. <https://doi.org/10.1186/s12909-024-06150-2>.

- 135 Hashmi OU, Chan N, de Vries CF, Gangi A, Jehanli L, Lip G. Artificial intelligence in radiology: trainees want more. *Clin Radiol*. 2023;78(4):e336–e341. <https://doi.org/10.1016/j.crad.2022.12.017>.
- 136 Sur J, Bose S, Khan F, Dewangan D, Sawriya E, Roul A. Knowledge, attitudes, and perceptions regarding the future of artificial intelligence in oral radiology in India: a survey. *Imaging Sci Dent*. 2020;50(3):193–198. <https://doi.org/10.5624/isd.2020.50.3.193>.
- 137 Farouk S, Osman AM, Awadallah SM, Abdelrahman AS. The added value of using artificial intelligence in adult chest X-rays for nodules and masses detection in daily radiology practice. *Egypt J Radiol Nucl Med*. 2023;54(1):142. <https://doi.org/10.1186/s43055-023-01093-y>.
- 138 Fujioka T, Kubota K, Hsu JF, et al. Examining the effectiveness of a deep learning-based computer-aided breast cancer detection system for breast ultrasound. *J Med Ultrason* (2001). 2023;50(4):511–520. <https://doi.org/10.1007/s10396-023-01332-9>.
- 139 Han X, He Y, Luo N, et al. The influence of artificial intelligence assistance on the diagnostic performance of CCTA for coronary stenosis for radiologists with different levels of experience. *Acta Radiol*. 2023;64(2):496–507. <https://doi.org/10.1177/02841851221089263>.
- 140 Kim T, Goh TS, Lee JS, Lee JH, Kim H, Jung ID. Transfer learning-based ensemble convolutional neural network for accelerated diagnosis of foot fractures. *Phys Eng Sci Med*. 2023;46(1):265–277. <https://doi.org/10.1007/s13246-023-01215-w>.
- 141 Kim JH, Han SG, Cho A, Shin HJ, Baek SE. Effect of deep learning-based assistive technology use on chest radiograph interpretation by emergency department physicians: a prospective interventional simulation-based study. *BMC Med Inform Decis Mak*. 2021;21(1):311. <https://doi.org/10.1186/s12911-021-01679-4>.
- 142 Nam JG, Kim M, Park J, et al. Development and validation of a deep learning algorithm detecting 10 common abnormalities on chest radiographs. *Eur Respir J*. 2021;57(5):2003061. <https://doi.org/10.1183/13993003.03061.2020>.
- 143 Liu Y, Chen C, Wang K, et al. The auxiliary diagnosis of thyroid echogenic foci based on a deep learning segmentation model: a two-center study. *Eur J Radiol*. 2023;167:111033. <https://doi.org/10.1016/j.ejrad.2023.111033>.
- 144 Wang DY, Liu SG, Ding J, et al. A deep learning model enhances clinicians' diagnostic accuracy to more than 96% for anterior cruciate ligament ruptures on magnetic resonance imaging. *Arthroscopy*. 2024;40(4):1197–1205. <https://doi.org/10.1016/j.arthro.2023.08.010>.
- 145 Yang W, Chen C, Yang Y, et al. Diagnostic performance of deep learning-based vessel extraction and stenosis detection on coronary computed tomography angiography for coronary artery disease: a multi-reader multi-case study. *Radiol Med*. 2023;128(3):307–315. <https://doi.org/10.1007/s11547-023-01606-9>.
- 146 Liu J, Zhao L, Han X, Ji H, Liu L, He W. Estimation of malignancy of pulmonary nodules at CT scans: effect of computer-aided diagnosis on diagnostic performance of radiologists. *Asia Pac J Clin Oncol*. 2021;17(3):216–221. <https://doi.org/10.1111/ajco.13362>.
- 147 Mehralivand S, Harmon SA, Shih JH, et al. Multicenter multi-reader evaluation of an artificial intelligence-based attention mapping system for the detection of prostate cancer with multiparametric MRI. *AJR Am J Roentgenol*. 2020;215(4):903–912. <https://doi.org/10.2214/AJR.19.22573>.
- 148 Hwang EJ, Kim H, Yoon SH, Goo JM, Park CM. Implementation of a deep learning-based computer-aided detection system for the interpretation of chest radiographs in patients suspected for COVID-19. *Korean J Radiol*. 2020;21(10):1150–1160. <https://doi.org/10.3348/kjr.2020.0536>.
- 149 Li C, Li J, Tan T, Chen K, Xu Y, Wu R. Application of ultrasonic dual-mode artificially intelligent architecture in assisting radiologists with different diagnostic levels on breast masses classification. *Diagn Interv Radiol*. 2021;27(3):315–322. <https://doi.org/10.5152/dir.2021.20018>.
- 150 van Winkel SL, Rodríguez-Ruiz A, Appelman L, et al. Impact of artificial intelligence support on accuracy and reading time in breast tomosynthesis image interpretation: a multi-reader multi-case study. *Eur Radiol*. 2021;31(11):8682–8691. <https://doi.org/10.1007/s00330-021-07992-w>.
- 151 Sim Y, Chung MJ, Kotter E, et al. Deep convolutional neural network-based software improves radiologist detection of malignant lung nodules on chest radiographs. *Radiology*. 2020;294(1):199–209. <https://doi.org/10.1148/radiol.2019182465>.
- 152 Zhang B, Jia C, Wu R, et al. Improving rib fracture detection accuracy and reading efficiency with deep learning-based detection software: a clinical evaluation. *Br J Radiol*. 2021;94(1118):20200870. <https://doi.org/10.1259/bjr.20200870>.
- 153 Bennani S, Regnard NE, Ventre J, et al. Using AI to improve radiologist performance in detection of abnormalities on chest radiographs. *Radiology*. 2023;309(3):e230860. <https://doi.org/10.1148/radiol.230860>.
- 154 Seah JCY, Tang CHM, Buchlak QD, et al. Effect of a comprehensive deep-learning model on the accuracy of chest x-ray interpretation by radiologists: a retrospective, multireader multicase study. *Lancet Digit Health*. 2021;3(8):e496–e506. [https://doi.org/10.1016/S2589-7500\(21\)00106-0](https://doi.org/10.1016/S2589-7500(21)00106-0).
- 155 Ahn JS, Ebrahimian S, McDermott S, et al. Association of artificial intelligence-aided chest radiograph interpretation with reader performance and efficiency. *JAMA Netw Open*. 2022;5(8):e2229289. <https://doi.org/10.1001/jamanetworkopen.2022.29289>.
- 156 Ha EJ, Lee JH, Lee DH, et al. Artificial intelligence model assisting thyroid nodule diagnosis and management: a multicenter diagnostic study. *J Clin Endocrinol Metab*. 2024;109(2):527–535. <https://doi.org/10.1210/clinem/dgaa503>.
- 157 Tong WJ, Wu SH, Cheng MQ, et al. Integration of artificial intelligence decision aids to reduce workload and enhance efficiency in thyroid nodule management. *JAMA Netw Open*. 2023;6(5):e2313674. <https://doi.org/10.1001/jamanetworkopen.2023.13674>.
- 158 Choi SY, Kim JH, Chung HS, Lim S, Kim EH, Choi A. Impact of a deep learning-based brain CT interpretation algorithm on clinical decision-making for intracranial hemorrhage in the emergency department. *Sci Rep*. 2024;14(1):22292. <https://doi.org/10.1038/s41598-024-73589-0>.
- 159 Novak A, Ather S, Gill A, et al. Evaluation of the impact of artificial intelligence-assisted image interpretation on the diagnostic performance of clinicians in identifying pneumothoraces on plain chest X-ray: a multi-case multi-reader study. *Emerg Med J*. 2024;41(10):602–609. <https://doi.org/10.1136/emerm-2023-213620>.
- 160 Liang F, Song Y, Huang X, et al. Assessing breast disease with deep learning model using bimodal bi-view ultrasound images and clinical information. *iScience*. 2024;27(7):110279. <https://doi.org/10.1016/j.isci.2024.110279>.
- 161 Lim DSW, Makmur A, Lei Z, et al. Improved productivity using deep learning-assisted reporting for lumbar spine MRI. *Radiology*. 2022;305(1):160–166. <https://doi.org/10.1148/radiol.220076>.
- 162 Bernstein MH, Atalay MK, Dibble EH, et al. Can incorrect artificial intelligence (AI) results impact radiologists, and if so, what can we do about it? A multi-reader pilot study of lung cancer detection with chest radiography. *Eur Radiol*. 2023;33(11):8263–8269. <https://doi.org/10.1007/s00330-023-09747-1>.
- 163 Hosny A, Parmar C, Quackenbush J, Schwartz LH, Aerts HJWL. Artificial intelligence in radiology. *Nat Rev Cancer*. 2018;18(8):500–510. <https://doi.org/10.1038/s41568-018-0016-5>.
- 164 Khalifa M, Albadawy M. AI in diagnostic imaging: revolutionising accuracy and efficiency. *Comput Methods Progr Biomed Update*. 2024;5:100146. <https://doi.org/10.1016/j.cmpbup.2024.100146>.
- 165 Morley J. "AI and the NHS: Is it the Silver Bullet That Will Improve the Health Service's Productivity?". United Kingdom (UK): Nuffield Trust; 2024. Guest blog.
- 166 Recht MP, Dewey M, Dreyer K, et al. Integrating artificial intelligence into the clinical practice of radiology: challenges and recommendations. *Eur Radiol*. 2020;30(6):3576–3584. <https://doi.org/10.1007/s00330-020-06672-5>.
- 167 Vo V, Chen G, Aquino YSJ, Carter SM, Do QN, Woode ME. Multi-stakeholder preferences for the use of artificial intelligence in healthcare: a systematic review and thematic analysis. *Soc Sci Med*. 2023;338:116357. <https://doi.org/10.1016/j.socscimed.2023.116357>.
- 168 Geis JR, Brady A, Wu CC, et al. Ethics of artificial intelligence in radiology: summary of the joint European and North American multisociety statement. *Insights Imaging*. 2019;10(1):101. <https://doi.org/10.1186/s13244-019-0785-8>.
- 169 European Union. Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 Laying Down Harmonised Rules on Artificial Intelligence and Amending Regulations. Europe: European Union; 2024.
- 170 National Institute for Health and Care Excellence (NICE). Artificial intelligence (AI)-derived software to help clinical decision making in stroke. Available from: <https://www.nice.org.uk/guidance/dg57/resources/artificial-intelligence-aided-software-to-help-clinical-decision-making-in-stroke-pdf-1053876693445>; 2024. Accessed September 9, 2024.
- 171 Karpachakis K, Morley J, Floridi L. A justifiable investment in AI for healthcare: aligning ambition with reality. *Minds Mach*. 2024;34:38.
- 172 Doherty G, McLaughlin L, Hughes C, McConnell J, Bond R, McFadden S. A scoping review of educational programmes on artificial intelligence (AI) available to medical imaging staff. *Radiography (Lond)*. 2024;30(2):474–482. <https://doi.org/10.1016/j.radi.2023.12.019>.